

Informe de experimento: results_validation_gpt-5.4_llm_only

Generado el 2026-09-15 23:37 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-5.4_llm_only (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_gpt-5.4_llm_only
```

- Reports leídos: 1 (report.rescored.json usado en 0; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: llm_only:cot, llm_only:structured, llm_only_forced:cot, llm_only_forced:structured
- Casos: case14
- Requests: generadas, N=5 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=8, parameterized=4, multistep=4, ambiguous=4
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-15 23:30
- Costo total: 0.3060 USD (20 items; costo medio 0.01530 USD/item)
- Items totales: 20; ok=20; con error=0; tokens totales=61079
- Solved global: 20.0% (4/20)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método llm_only:cot

System prompt —734 caracteres; strategy=cot, forced=False, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- If you cannot complete the calculation, set `converged` to false and output the structure you can provide (an empty array is acceptable).

Reasoning mode: you MAY write a short step-by-step reasoning in plain text before the answer. The final JSON object must be the LAST thing
↪ in the response, inside a single ```json fenced block, and it must be the only JSON object in the response. Do not use curly braces in
↪ the reasoning text.

User prompt (primer item) —6326 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	type	zone	in_service
0	1	b	1.0	True
1	2	b	1.0	True
2	3	b	1.0	True
3	4	b	1.0	True
4	5	b	1.0	True
5	6	b	1.0	True
6	7	b	1.0	True
7	8	b	1.0	True
8	9	b	1.0	True
9	10	b	1.0	True
10	11	b	1.0	True

... (3 líneas omitidas de 15)

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
0	0	1.06	0.0	True

Line Data (Line Table: net.line)

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True
1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vkr_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min
0	↪	tap_max	tap_step_percent	in_service									
↪	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		2.2	True									
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		3.1	True									
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		6.8	True									
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Reasoning Instructions

Before producing the final JSON, reason step by step:

1. Which buses does the request refer to (display ids, slack bus, load and generator buses)?
 2. Which lines/branches are involved (from_bus-to_bus, line_id convention)?
 3. Which quantities and units are asked for (MW, Mvar, p.u., degrees, % loading)?
 4. What changes with respect to the base case, and which power balance must hold?
- Write the reasoning first, then the final JSON as the last part of the response.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
 - For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

Método llm_only:structured

System prompt —439 caracteres; strategy=structured, forced=False, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- If you cannot complete the calculation, set `converged` to false and output the structure you can provide (an empty array is acceptable).

User prompt (primer item) —5831 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	type	zone	in_service
0	1	b	1.0	True
1	2	b	1.0	True
2	3	b	1.0	True
3	4	b	1.0	True
4	5	b	1.0	True
5	6	b	1.0	True
6	7	b	1.0	True
7	8	b	1.0	True
8	9	b	1.0	True
9	10	b	1.0	True
10	11	b	1.0	True
...	(3 líneas omitidas de 15)			

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
0	0	1.06	0.0	True

Line Data (Line Table: net.line)

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True

1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vkr_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min
	↪ tap_max	tap_step_percent		in_service									
0	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		2.2	True									
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		3.1	True									
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		6.8	True									
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
 - For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,

  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],

  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],

  "total_generation_mw": 0.0,

  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

Método llm_only_forced:cot

System prompt —860 caracteres; strategy=cot, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every ↪ array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

Reasoning mode: you MAY write a short step-by-step reasoning in plain text before the answer. The final JSON object must be the LAST thing ↪ in the response, inside a single ```json fenced block, and it must be the only JSON object in the response. Do not use curly braces in ↪ the reasoning text.

User prompt (primer item) —6326 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	type	zone	in_service
0	1	b	1.0	True
1	2	b	1.0	True
2	3	b	1.0	True
3	4	b	1.0	True
4	5	b	1.0	True
5	6	b	1.0	True
6	7	b	1.0	True
7	8	b	1.0	True
8	9	b	1.0	True
9	10	b	1.0	True
10	11	b	1.0	True

... (3 líneas omitidas de 15)

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
0	0	1.06	0.0	True

Line Data (Line Table: net.line)

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True
1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vkr_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min
↪	tap_max	tap_step_percent	in_service										
0	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		2.2	True									
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		3.1	True									
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN
↪	NaN		6.8	True									
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN
↪	NaN		NaN	True									

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Reasoning Instructions

Before producing the final JSON, reason step by step:

1. Which buses does the request refer to (display ids, slack bus, load and generator buses)?
 2. Which lines/branches are involved (from_bus-to_bus, line_id convention)?
 3. Which quantities and units are asked for (MW, Mvar, p.u., degrees, % loading)?
 4. What changes with respect to the base case, and which power balance must hold?
- Write the reasoning first, then the final JSON as the last part of the response.

```

## Output Requirements
Please calculate and return the following results (JSON format):

1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
2) Active power (MW) and load factor (%) at the beginning of each branch
- For lines: `line_id` uses the row index of `net.line`
- For transformers: `line_id` uses the row index of 100000 + `net.trafo`
3) Total generation (MW), total load (MW), total losses (MW)
4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):
{
  "converged": true/false,
  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],
  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

```

Método llm_only_forced:structured

System prompt —565 caracteres; strategy=structured, forced=True, sin herramientas

You are a power system analysis expert. You need to calculate AC power flow results based on test system data provided by the user.

Important Constraints:

- You must strictly output JSON (no extra text, no Markdown blocks).
- Do not output NaN/Infinity; all values must be finite real numbers.
- You must compute and report your best numerical estimate for every bus voltage and every branch flow. `converged` must be true and every array must be complete (one entry per bus and per branch). Do not leave arrays empty and do not state that you cannot compute.

User prompt (primer item) —5831 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

You are a power systems expert. Based on the following IEEE case14 test system data, calculate the AC power flow results.

System Data (from pandapower network tables)

Node Data (Bus Table: net.bus)

	name	type	zone	in_service
0	1	b	1.0	True
1	2	b	1.0	True
2	3	b	1.0	True
3	4	b	1.0	True
4	5	b	1.0	True
5	6	b	1.0	True
6	7	b	1.0	True
7	8	b	1.0	True
8	9	b	1.0	True
9	10	b	1.0	True
10	11	b	1.0	True
...	(3 líneas omitidas de 15)			

Load Data (Load Table: net.load)

	bus	p_mw	q_mvar	in_service
0	1	21.473798	12.567615	True
1	2	85.726702	17.290948	True
2	3	45.568126	-3.717902	True
3	4	6.887322	1.449962	True
4	5	11.740497	7.861940	True
5	8	32.229269	18.135792	True
6	9	9.714580	6.260507	True
7	10	3.424253	1.761044	True
8	11	6.539657	1.715320	True
9	12	14.757610	6.340306	True
10	13	15.783969	5.296634	True

Generator Data (Generator Table: net.gen)

	bus	p_mw	vm_pu	min_p_mw	max_p_mw	min_q_mvar	max_q_mvar	in_service
0	1	41.63323	1.045	0.0	140.0	-40.0	50.0	True
1	2	0.00000	1.010	0.0	100.0	0.0	40.0	True
2	5	0.00000	1.070	0.0	100.0	-6.0	24.0	True
3	7	0.00000	1.090	0.0	100.0	-6.0	24.0	True

Balanced Node/External Grid (External Grid Table: net.ext_grid)

	bus	vm_pu	va_degree	in_service
0	0	1.06	0.0	True

Line Data (Line Table: net.line)

	from_bus	to_bus	length_km	r_ohm_per_km	x_ohm_per_km	c_nf_per_km	max_i_ka	df	parallel	in_service
0	0	1	1.0	3.532005	10.783732	768.484773	42.339020	1.0	1	True
1	0	4	1.0	9.846967	40.649040	716.088084	42.339020	1.0	1	True
2	1	2	1.0	8.563928	36.080033	637.493051	42.339020	1.0	1	True
3	1	3	1.0	10.590547	32.134320	494.857619	42.339020	1.0	1	True
4	1	4	1.0	10.379138	31.689630	503.590401	42.339020	1.0	1	True
5	2	3	1.0	12.212573	31.170217	186.299339	42.339020	1.0	1	True
6	3	4	1.0	2.433038	7.674548	0.000000	42.339020	1.0	1	True
7	5	10	1.0	0.000041	0.000086	0.000000	27479.652235	1.0	1	True
8	5	11	1.0	0.000053	0.000111	0.000000	27479.652235	1.0	1	True
9	5	12	1.0	0.000029	0.000056	0.000000	27479.652235	1.0	1	True
10	8	9	1.0	0.000014	0.000037	0.000000	27479.652235	1.0	1	True

... (4 líneas omitidas de 16)

Transformer Data (Transformer Table: net.trafo)

	hv_bus	lv_bus	sn_mva	vn_hv_kv	vn_lv_kv	vk_percent	vkr_percent	pfe_kw	i0_percent	shift_degree	tap_side	tap_neutral	tap_min	
0	↪ tap_max	3	6	9900.0	135.0	14.000	2070.288	0.0	0.0	0.0	0.0	hv	0.0	NaN
	↪ tap_step_percent	NaN	2.2	True										
1	3	8	9900.0	135.0	0.208	5506.182	0.0	0.0	0.0	0.0	hv	0.0	NaN	
	↪ tap_max	NaN	3.1	True										
2	4	5	9900.0	135.0	0.208	2494.998	0.0	0.0	0.0	0.0	hv	0.0	NaN	
	↪ tap_max	NaN	6.8	True										
3	6	7	9900.0	14.0	12.000	1743.885	0.0	0.0	0.0	0.0	None	NaN	NaN	
	↪ tap_max	NaN	NaN	True										
4	6	8	9900.0	14.0	0.208	1089.099	0.0	0.0	0.0	0.0	None	NaN	NaN	
	↪ tap_max	NaN	NaN	True										

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Please calculate and return the following results (JSON format):

- 1) Voltage magnitude (p.u.) and phase angle (degrees) for each node
- 2) Active power (MW) and load factor (%) at the beginning of each branch
 - For lines: `line\_id` uses the row index of `net.line`
 - For transformers: `line\_id` uses the row index of 100000 + `net.trafo`
- 3) Total generation (MW), total load (MW), total losses (MW)
- 4) Are there any voltage overruns (<0.95 or >1.05 p.u.) or line overruns (>100%)?

Please strictly follow the following JSON schema output (all fields are complete and of correct type):

```
{
  "converged": true/false,
  "bus_voltages": [{"bus_id": 1, "vm_pu": 1.0, "va_deg": 0.0}, ...],
  "line_flows": [{"line_id": 0, "p_from_mw": 0.0, "loading_percent": 0.0}, ...],
  "total_generation_mw": 0.0,
  "total_load_mw": 0.0,
  "total_loss_mw": 0.0
}
```

[Important] bus_id must use the `name` column of net.bus (the usual IEEE/MATPOWER 1..N numbering), not the row index of net.bus.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion` **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status

(convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	\$
llm_only:cot	case14	5	–	–	–	0.0 (0/5)	0.0 (0/5)	–	3.3	–	n/a	2944	–	60.0 (3/5)	0.0635
llm_only:structured	case14	5	–	–	–	0.0 (0/5)	0.0 (0/5)	–	0.0	–	n/a	2390	–	80.0 (4/5)	0.0330
llm_only_forced:cot	case14	5	–	2.18e-02	5.01	40.0 (2/5)	100.0 (5/5)	–	0.0	–	n/a	3696	–	0.0 (0/5)	0.1183
llm_only_forced:structured	case14	5	–	6.01e-03	4.80	40.0 (2/5)	100.0 (5/5)	–	0.0	–	n/a	3186	–	0.0 (0/5)	0.0911

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
llm_only:cot	n/a	n/a	n/a	n/a
llm_only:structured	n/a	n/a	n/a	n/a
llm_only_forced:cot	n/a	n/a	n/a	n/a
llm_only_forced:structured	n/a	n/a	n/a	n/a

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
llm_only:cot	0.0 (0/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
llm_only:structured	0.0 (0/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
llm_only_forced:cot	50.0 (1/2)	100.0 (1/1)	0.0 (0/1)	0.0 (0/1)
llm_only_forced:structured	50.0 (1/2)	100.0 (1/1)	0.0 (0/1)	0.0 (0/1)

Tipos de error de formulación (conteo de items)

(ningún item con *formulation_error_type*)

Errores de ejecución por método (conteo de items con error)

(ningún item con *error*)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método llm_only:cot

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
Reasoning
1. Bus ids requested in output use the net.bus name column, so buses are 1 through 14. The external grid is connected at row index bus 0,
↳ which corresponds to display id 1 and is the slack bus at 1.06 p.u. and 0 degrees. Generator buses from net.gen are row indices 1, 2,
↳ 5, 7, corresponding to display ids 2, 3, 6, 8. Load buses are row indices 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13, corresponding to display
↳ ids 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14.

2. The line to disconnect is the branch between bus ten and bus eleven in display numbering. In zero based row indices that is between row
↳ index 9 and 10. In net.line, the branch with from_bus 9 and to_bus 10 is line row 12, so line_id 12 is opened. Other branches remain in
↳ service. Transformer ids in output use 100000 plus transformer r ...[1171 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=--; tokens=3056; cost_usd=0.0146; wall=6.8 s
```

Ejemplo: no resuelto

```
request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
Step 1
Bus display ids are the net.bus name values 1 through 14, while the internal row indices are 0 through 13. The slack bus is external grid
↳ at row index 0, which corresponds to bus id 1. Generator buses are row indices 1, 2, 5, 7 corresponding to bus ids 2, 3, 6, 8. Load
↳ buses are at row indices 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13 corresponding to bus ids 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14.

Step 2
The requested outages are stated as bus 6 to bus 13 and bus 12 to bus 13 in display numbering. In the line table, display bus 6 is row
↳ index 5, display bus 13 is row index 12, and display bus 12 is row index 11. The corresponding existing lines are line 9 with from bus
↳ 5 to 12 internal, meaning display buses 6 to 13, and line 13 with from bus 11 to 12 internal, meaning display buses 12 to ...[1217 chars
↳ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=16.7 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=3115; cost_usd=0.0149; wall=7.6 s
```

Ejemplo: primer item

```
request (case14-parameterized-001-s0, case14, difficulty=parameterized, seed=827308000): Load the IEEE 14-bus system and run an N-1
↳ contingency analysis ranked by min voltage and report the 3 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 3, "criteria": "min_voltage"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
Buses use display ids from net.bus name 1 through 14. Slack is external grid at row bus 0 which corresponds to display bus 1. Generator
↳ buses are display buses 2, 3, 6, 8 from net.gen row buses 1, 2, 5, 7 plus slack at bus 1. Load buses are display buses 2, 3, 4, 5, 6,
↳ 9, 10, 11, 12, 13, 14 from net.load row buses 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13.

Branches are all net.line rows with line_id equal to the line table index and transformer branches with line_id equal to 100000 plus
↳ transformer table index. Line endpoints in display bus ids are 1-2, 1-5, 2-3, 2-4, 2-5, 3-4, 4-5, 6-11, 6-12, 6-13, 9-10, 9-14, 10-11,
↳ 12-13, 13-14. Transformer endpoints in display ids are 4-7, 4-9, 5-6, 7-8, 7-9.

Requested quantities are bus voltage magnitude in per unit and angle in degrees, branch sending-e ...[684 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=3001; cost_usd=0.0136; wall=5.6 s
```

Método llm_only:structured

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":false,"bus_voltages":[],"line_flows":[],"total_generation_mw":0.0,"total_load_mw":0.0,"total_loss_mw":0.0}
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2376; cost_usd=0.0066; wall=1.3 s
```

Ejemplo: no resuelto

```
request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":false,"bus_voltages":[],"line_flows":[],"total_generation_mw":0.0,"total_load_mw":0.0,"total_loss_mw":0.0}
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2417; cost_usd=0.0067; wall=2.3 s
```

Ejemplo: primer item

```
request (case14-parameterized-001-s0, case14, difficulty=parameterized, seed=827308000): Load the IEEE 14-bus system and run an N-1
↳ contingency analysis ranked by min voltage and report the 3 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 3, "criteria": "min_voltage"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
```

```

--- Respuesta final ---
{"converged":false,"bus_voltages":[],"line_flows":[],"total_generation_mw":0.0,"total_load_mw":0.0,"total_loss_mw":0.0}
--- Scoring ---
formulation=n/a (exact=None); V_MAE=--; F_MAE=--; faithful=0.0 %; solved=False (convergence_mismatch);
↳ flags=abstained,abstained_on_solvable; error=-; tokens=2390; cost_usd=0.0066; wall=1.7 s

```

Método llm_only_forced:cot

Ejemplo: resuelto (solved=True)

```

request (case14-parameterized-001-s0, case14, difficulty=parameterized, seed=827308000): Load the IEEE 14-bus system and run an N-1
↳ contingency analysis ranked by min voltage and report the 3 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 3, "criteria": "min_voltage"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
 "converged": true,
 "bus_voltages": [
 {"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0},
 {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.98},
 {"bus_id": 3, "vm_pu": 1.01, "va_deg": -12.72},
 {"bus_id": 4, "vm_pu": 1.017, "va_deg": -10.33},
 {"bus_id": 5, "vm_pu": 1.02, "va_deg": -8.78},
 {"bus_id": 6, "vm_pu": 1.07, "va_deg": -14.22},
 {"bus_id": 7, "vm_pu": 1.062, "va_deg": -13.37},
 {"bus_id": 8, "vm_pu": 1.09, "va_deg": -13.36},
 {"bus_id": 9, "vm_pu": 1.056, "va_deg": -14.94},
 {"bus_id": 10, "vm_pu": 1.051, "va_deg": -15.1},
 {"bus_id": 11, "vm_pu": 1.057, "va_deg": -14.79},
 {"bus_id": 12, "vm_pu": 1.055, "va_deg": -15.08},
 {"bus_id": 13, "vm_pu": 1.05, "va_deg": -15.16},
 {"bus_id": 14, "vm_pu": 1.036, "va_deg": -16.04}
],
 "line_fl ...[1405 chars más]
}
--- Scoring ---
formulation=n/a (exact=None); V_MAE=3.49e-04; F_MAE=4.00; faithful=0.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3586;
↳ cost_usd=0.0217; wall=0.0 s

```

### Ejemplo: no resuelto

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
Bus ids by name are 1 to 14. Slack bus is display bus 1 and corresponds to ext_grid at row-index bus 0. PV generator buses are display
↳ buses 2, 3, 6, 8 corresponding to row-index buses 1, 2, 5, 7. Load buses are display buses 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14 from
↳ the load table row-index mapping plus the pure PQ buses without generation control.

The branch ids for lines use net.line row index 0 to 14. The requested outage is the line between bus ten and bus eleven in display
↳ numbering, which corresponds to display buses 10 and 11. In the provided net.line table there is no direct 10 to 11 line; the closest
↳ relevant line is row 12 connecting row-index buses 9 to 10, which are display buses 10 to 11. So line_id 12 is disconnected.
↳ Transformer ids use 100000 plus trafo row index, so 1000 ...[3154 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=1.16e-02; F_MAE=9.40; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3817;
↳ cost_usd=0.0256; wall=0.0 s

```

### Ejemplo: primer item

```

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
```json
{
  "converged": true,
  "bus_voltages": [
    {"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0},
    {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.98},
    {"bus_id": 3, "vm_pu": 1.01, "va_deg": -12.72},
    {"bus_id": 4, "vm_pu": 1.017, "va_deg": -10.33},
    {"bus_id": 5, "vm_pu": 1.019, "va_deg": -8.78},
    {"bus_id": 6, "vm_pu": 1.07, "va_deg": -14.22},
    {"bus_id": 7, "vm_pu": 1.061, "va_deg": -13.37},
    {"bus_id": 8, "vm_pu": 1.09, "va_deg": -13.36},
    {"bus_id": 9, "vm_pu": 1.056, "va_deg": -14.94},

```

```

{"bus_id": 10, "vm_pu": 1.051, "va_deg": -15.1},
{"bus_id": 11, "vm_pu": 1.057, "va_deg": -14.79},
{"bus_id": 12, "vm_pu": 1.055, "va_deg": -15.08},
{"bus_id": 13, "vm_pu": 1.05, "va_deg": -15.16},
{"bus_id": 14, "vm_pu": 1.036, "va_deg": -16.04}
],
"line_f ...[1422 chars más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.02e-02; F_MAE=8.00; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3561;
↪ cost_usd=0.0216; wall=0.0 s

```

Método llm_only_forced:structured

Ejemplo: resuelto (solved=True)

```

request (case14-parameterized-001-s0, case14, difficulty=parameterized, seed=827308000): Load the IEEE 14-bus system and run an N-1
↪ contingency analysis ranked by min voltage and report the 3 worst outages.
intended_calls: load_case({"case_name": "case14"}) -> run_n1_contingency({"top_k": 3, "criteria": "min_voltage"})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0},{"bus_id":2,"vm_pu":1.045,"va_deg":-4.98},{"bus_id":3,"vm_pu":1.01
, "va_deg":-12.72},{"bus_id":4,"vm_pu":1.017,"va_deg":-10.33},{"bus_id":5,"vm_pu":1.02,"va_deg":-8.78},{"bus_id":6,"vm_pu":1.07,"va_deg"
↪ :-14.22},{"bus_id":7,"vm_pu":1.062,"va_deg":-13.37},{"bus_id":8,"vm_pu":1.09,"va_deg":-13.36},{"bus_id":9,"vm_pu":1.056,"va_deg":-14.94
↪ },{"bus_id":10,"vm_pu":1.051,"va_deg":-15.1},{"bus_id":11,"vm_pu":1.057,"va_deg":-14.79},{"bus_id":12,"vm_pu":1.055,"va_deg":-15.08},{"
↪ bus_id":13,"vm_pu":1.05,"va_deg":-15.16},{"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows":[{"line_id":0,"p_from_mw":156.88,"lo
↪ ading_percent":77.4},{"line_id":1,"p_from_mw":75.51,"loading_percent":45.1},{"line_id":2,"p_from_mw":73.24,"loading_perce ...[1038 chars
↪ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=4.20e-04; F_MAE=2.66; faithful=0.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=3166;
↪ cost_usd=0.0176; wall=0.0 s

```

Ejemplo: no resuelto

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↪ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0},{"bus_id":2,"vm_pu":1.045,"va_deg":-4.98},{"bus_id":3,"vm_pu":1.01
, "va_deg":-12.72},{"bus_id":4,"vm_pu":1.017,"va_deg":-10.33},{"bus_id":5,"vm_pu":1.019,"va_deg":-8.78},{"bus_id":6,"vm_pu":1.07,"va_deg"
↪ ":-14.22},{"bus_id":7,"vm_pu":1.062,"va_deg":-13.37},{"bus_id":8,"vm_pu":1.09,"va_deg":-13.36},{"bus_id":9,"vm_pu":1.056,"va_deg":-14.9
↪ 4},{"bus_id":10,"vm_pu":1.051,"va_deg":-15.1},{"bus_id":11,"vm_pu":1.057,"va_deg":-14.79},{"bus_id":12,"vm_pu":1.055,"va_deg":-15.08},{
↪ "bus_id":13,"vm_pu":1.05,"va_deg":-15.16},{"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows":[{"line_id":0,"p_from_mw":147.2,"lo
↪ ading_percent":52.4},{"line_id":1,"p_from_mw":71.6,"loading_percent":27.8},{"line_id":2,"p_from_mw":70.1,"loading_percent ...[1014 chars
↪ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=3.14e-03; F_MAE=3.19; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3132;
↪ cost_usd=0.0175; wall=0.0 s

```

Ejemplo: primer item

```

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↪ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↪ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↪ "to_bus": 13}) -> run_powerflow({})
executed_calls: (sin etapa de tools)
fuente de la traza: archivo de traza completo
(sin rondas de herramientas: una sola llamada al LLM)
--- Respuesta final ---
{"converged":true,"bus_voltages":[{"bus_id":1,"vm_pu":1.06,"va_deg":0.0},{"bus_id":2,"vm_pu":1.045,"va_deg":-4.98},{"bus_id":3,"vm_pu":1.01
, "va_deg":-12.72},{"bus_id":4,"vm_pu":1.017,"va_deg":-10.33},{"bus_id":5,"vm_pu":1.019,"va_deg":-8.78},{"bus_id":6,"vm_pu":1.07,"va_deg"
↪ ":-14.22},{"bus_id":7,"vm_pu":1.062,"va_deg":-13.37},{"bus_id":8,"vm_pu":1.09,"va_deg":-13.36},{"bus_id":9,"vm_pu":1.056,"va_deg":-14.9
↪ 4},{"bus_id":10,"vm_pu":1.051,"va_deg":-15.1},{"bus_id":11,"vm_pu":1.057,"va_deg":-14.79},{"bus_id":12,"vm_pu":1.055,"va_deg":-15.07},{
↪ "bus_id":13,"vm_pu":1.05,"va_deg":-15.16},{"bus_id":14,"vm_pu":1.036,"va_deg":-16.04}], "line_flows":[{"line_id":0,"p_from_mw":156.88,"l
↪ oading_percent":100.2},{"line_id":1,"p_from_mw":75.51,"loading_percent":72.3},{"line_id":2,"p_from_mw":73.24,"loading_per ...[1036 chars
↪ más]
--- Scoring ---
formulation=n/a (exact=None); V_MAE=2.02e-02; F_MAE=10.84; faithful=0.0 %; solved=False (voltage_error); flags=-; error=-; tokens=3140;
↪ cost_usd=0.0175; wall=0.0 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- LLM-only: V_MAE entre 6.01e-03 (llm_only_forced:structured/case14) y 2.18e-02 (llm_only_forced:cot/case14) p.u. sobre los items que devolvieron números.

- 35.0 % (7/20) de abstención en LLM-only (respuesta con `converged=false` o arrays vacíos).
- Abstención con cláusula de escape: 70.0 % (7/10); sin cláusula (forced): 0.0 % (0/10).
- En LLM-only, 10 items no resueltos por `convergence_mismatch` y 6 por `voltage_error`.
- Ningún item terminó con `error`.
- Solved más alto: `llm_only_forced:structured` 40.0 % (2/5); más bajo: `llm_only:cot` 0.0 % (0/5).
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- Trazabilidad de números (Faith.) media en LLM-only: 0.8 % (sin tools, todo número es no trazable salvo los copiados de la request).
- Costo total 0.3060 USD; el método más caro fue `llm_only_forced:cot` con 0.1183 USD (39 %).

6. Archivos

Rutas absolutas bajo `results_validation_gpt-5.4_llm_only. report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```
report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/report.json
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/scoreboard_per_case.json
traces (30 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_llm_only/traces
```